Literature-Implied Status: Uncountable $C(K)$ Calkin Algebra

FA/Banach run `fa_banach_001`

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Status

This is a literature-implied answer packet, not an original proof from the run. Problem 4 in Motakis–Puglisi–Tolias, arXiv:1711.01340, asks whether some Banach-space Calkin algebra is isomorphic, as a Banach algebra, to $C(K)$ for an uncountable compact topological space K.

Motakis, arXiv:2110.10868, proves the stronger metric compact case: for every compact metric space K, there is a Banach space X whose Calkin algebra is homomorphically isometric to $C(K)$. Taking $K = [0, 1]$ or the Cantor set answers the 2017 problem affirmatively.

Original Problem

The original source is:

> P. Motakis, D. Puglisi, A. Tolias, *Algebras of diagonal operators of the form scalar-plus-compact are Calkin algebras*, arXiv:1711.01340.

Problem 4 asks for a Calkin algebra isomorphic to $C(K)$ with K an uncountable compact topological space.

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Problem 4. Does there exist a Banach space the Calkin algebra of which is isomorphic, as a Banach algebra to $C(K)$ for an uncountable compact topological space K?

Remark 8.8. In a personal communication with the first author T. Kania asked whether the unitization of Schreier space from [Schr] endowed with coordinate-wise multiplication with respect to its standard unconditional basis is a Calkin algebra

Supporting Theorem

The supporting source is:

> P. Motakis, *Separable Spaces of Continuous Functions as Calkin Algebras*, arXiv:2110.10868.

Theorem A states that, for every compact metric space K, there exists an Argyros–Haydon-type Bourgain–Delbaen space $\mathfrak{X}_{C(K)}$ such that

$$\mathrm{Cal}(\mathfrak{X}_{C(K)}) \cong C(K)$$

as Banach algebras; after an equivalent renorming the isomorphism is isometric.

sired result. With regards to the first step of the three-step process towards constructing a $C(K)$ Calkin algebra, normal operators on ℓ_2 are the natural candidates. Indeed, by the spectral theorem the C^*-algebra generated by a normal operator T is $C(\sigma(T))$. This is in fact fairly straightforward whenever T is diagonal with respect to some orthonormal basis and thus its norm is the supremum of its diagonal entries. Going through this exercise clarifies that on any space with an unconditional basis any family of diagonal operators generates a $C(K)$ Banach algebra. This supports the conclusion that the class $\mathcal{C}$ in the first step needs to be one consisting of sufficiently many diagonal operators that capture the information of the compact space K.

Theorem A. Let K be a compact metric space. There exists an Argyros-Haydon-type Bourgain-Delbaen $\mathscr{L}_\infty$-space $\mathfrak{X}_{C(K)}$ with a conditional Schauder basis $(d_\gamma)_{\gamma\in\Gamma}$ that satisfies the following properties.

(a) Every bounded linear operator $T : \mathfrak{X}_{C(K)} \to \mathfrak{X}_{C(K)}$ can be written in the form $T = D + A$ where D is diagonal bounded linear operator and A is a compact linear operator.
(b) There exists a Banach algebra isomorphism $\Psi : C(K) \to \mathcal{C}al(\mathfrak{X}_{C(K)})$.
(c) The space $\mathfrak{X}_{C(K)}$ admits an equivalent norm with respect to which Ψ is an isometry. In particular, this space's Calkin algebra is homomorphically isometric to $C(K)$.

Implication

Let $K = [0, 1]$. Then K is compact, metric, and uncountable. Applying Motakis's theorem yields a Banach space X with

$$\mathrm{Cal}(X) \cong C([0, 1])$$

as Banach algebras. This is precisely the requested type of Calkin algebra in Problem 4 of arXiv:1711.01340.

Scope

The classification is literature-implied because the supporting paper proves the needed general theorem without, as far as checked, explicitly naming Problem 4 of arXiv:1711.01340. The result answers the uncountable compact *space* question, but it does not answer the later problem asking for a nonseparable $C(K)$ Calkin algebra.

Review Recommendation

Verify the reading of the original word "uncountable": it modifies the compact space K, not the density of $C(K)$. Under that reading the compact metric examples above are enough.